A Pedagogical Guide to Winding-Free Exact Diagonalization for Quantum Spin Ice

Implementation note for the `twist_qsi_demo` repository

July 22, 2026

Abstract

This note gives an operator-level definition of winding-free exact diagonalization. A small periodic pyrochlore cluster admits spin-exchange sequences that return to the same wrapped configuration only after crossing a supercell boundary. The protocol inserts an explicit transport character in every microscopic exchange matrix element, diagonalizes the resulting Hamiltonian at the physical coupling, and extracts the zero-character Fourier component of the isolated band connected to the ice-rule subspace. We define the ice projector, the sourced Hamiltonian, its exact spectral projector, and the polar pullback separately. This makes clear both why the production calculation is nonperturbative and where its finite-grid and isolated-band assumptions enter. Low-order path enumeration is used only to verify the transport interpretation.

Contents

1 The finite-torus problem and the ice projector

The local-axis XYZ model is

$$H = H_0 + V = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} \left(S_i^+ S_j^- + S_i^- S_j^+ \right) + J_{\pm\pm} \sum_{\langle ij \rangle} \left(S_i^+ S_j^+ + S_i^- S_j^- \right). \tag{1}$$

The QMC benchmark and analytic loop scales below use $J_{\pm\pm} = 0$, but the exact source construction and cubic-16 implementation allow finite $J_{\pm\pm}$. For each tetrahedron t, define its Ising charge

$$\widehat{Q}_t = \sum_{i \in t} S_i^z. \tag{2}$$

All $\widehat{Q}_t$ commute. The reference space is therefore specified without choosing or counting basis states. For an S^z basis state $|s\rangle$, the local factor acts as

$$\mathbf{1}_{\{0\}}(\widehat{Q}_t)|s\rangle = \begin{cases} |s\rangle, & Q_t(s) = 0, \\ 0, & Q_t(s) \neq 0. \end{cases} \tag{3}$$

Thus $\mathbf{1}_{\{0\}}(\widehat{Q}_t)$ is just the zero-charge projector on tetrahedron t. Because a spin-1/2 tetrahedron has $Q_t \in \{-2, -1, 0, 1, 2\}$, it can also be evaluated as

$$\mathbf{1}_{\{0\}}(\widehat{Q}_t) = (1 - \widehat{Q}_t^2)(1 - \widehat{Q}_t^2/4). \tag{4}$$

The global projector is the product of these commuting local conditions:

$$P_{\text{ice}} = \prod_t \mathbf{1}_{\{0\}}(\widehat{Q}_t) = \sum_{\substack{s:\, Q_t(s)=0 \\ \text{for every } t}} |s\rangle\langle s|. \tag{5}$$

It obeys $P_{\text{ice}}^\dagger = P_{\text{ice}}$ and $P_{\text{ice}}^2 = P_{\text{ice}}$, so it is the orthogonal projector onto the subspace of states satisfying the two-in–two-out rule on every tetrahedron. Its rank depends on the cluster, but no step in the protocol is defined by that rank.

In the thermodynamic lattice, the first off-diagonal process within the ice-rule subspace is the six-spin ring exchange around a contractible pyrochlore hexagon, with scale

$$g_6 = 12 \frac{|J_{\pm}|^3}{J_{zz}^2}. \tag{6}$$

A periodic finite cluster has an additional possibility. A sequence may return to the same finite-cell spin configuration while transporting spin through a nonzero supercell image. Cubic-16 consequently admits ice-preserving four-loops with scale

$$g_4 = 4\frac{J_\pm^2}{J_{zz}}. \tag{7}$$

This is a finite-topology artifact, not a local bulk plaquette. Because it is lower order than g_6, ordinary periodic exact diagonalization can put the low-temperature peak at the wrong scale even when the microscopic Hamiltonian itself is correct.

The objective is deliberately specific: remove nonzero covering-lattice transport from the exact band continuously connected to the ice-rule subspace, while retaining its winding-free ($\boldsymbol{q} = 0$) component. It does not assert that the resulting finite cluster is already a bulk surrogate. Character resolution, band isolation, cluster size, thermodynamics, and external QMC remain independent validation gates.

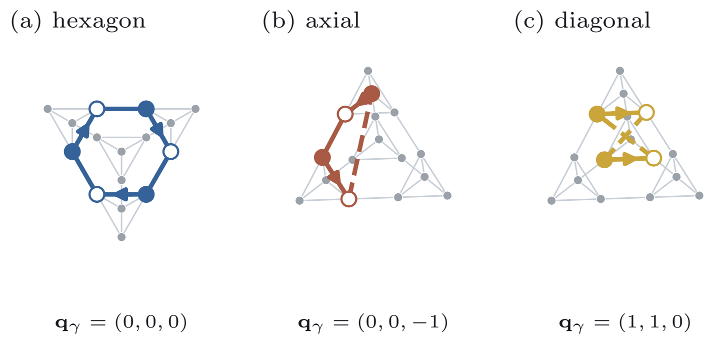

Figure 1: The actual cubic-16 pyrochlore network without an external cubic frame. The two four-loop classes close through periodic bonds (dashed) and carry finite $\boldsymbol{q}_\gamma = 2\sum_\ell \Delta\boldsymbol{p}_\ell$, shown as the dark resultant of the colored spin-transfer arrows. The contractible ring is viewed normal to its plane, making its D_6 geometry and $\boldsymbol{q}_\gamma = 0$ explicit.

2 The microscopic source is an explicit phase

Each site has a home-cell position $\boldsymbol{r}_i$, and the rows of L are the three supercell vectors. Each nearest-neighbor bond stores an integer image $\boldsymbol{n}_{ij}$. Its displacement in the covering lattice is therefore

$$\boldsymbol{d}_{ij} = \boldsymbol{r}_j - \boldsymbol{r}_i - \boldsymbol{n}_{ij}^T L, \tag{8}$$

from site i to the chosen image of site j. For a pair flip we also need $\boldsymbol{c}_{ij} = \boldsymbol{r}_i + \boldsymbol{r}_j - \boldsymbol{n}_{ij}^T L$. This is fixed by polarization bookkeeping. With $\widetilde{\boldsymbol{r}}_j = \boldsymbol{r}_j - \boldsymbol{n}_{ij}^T L$, raising both spins changes the lifted polarization by

$$\Delta\boldsymbol{p}(S_i^+ S_j^+) = \boldsymbol{r}_i + \widetilde{\boldsymbol{r}}_j = \boldsymbol{c}_{ij}, \tag{9}$$

while $S_i^+ S_j^-$ changes it by $\boldsymbol{r}_i - \widetilde{\boldsymbol{r}}_j = -\boldsymbol{d}_{ij}$. The source on every move is the single generating factor $e^{-2i\boldsymbol{\theta}\cdot\Delta\boldsymbol{p}}$.

The implemented source family is

$$H(\boldsymbol{\theta}) = H_0 - J_\pm \sum_{\langle ij\rangle} \left[e^{+2i\boldsymbol{\theta}\cdot\boldsymbol{d}_{ij}} S_i^+ S_j^- + e^{-2i\boldsymbol{\theta}\cdot\boldsymbol{d}_{ij}} S_i^- S_j^+ \right] + J_{\pm\pm} \sum_{\langle ij\rangle} \left[e^{-2i\boldsymbol{\theta}\cdot\boldsymbol{c}_{ij}} S_i^+ S_j^+ + e^{+2i\boldsymbol{\theta}\cdot\boldsymbol{c}_{ij}} S_i^- S_j^- \right]. \tag{10}$$

Thus "attaching a source" means modifying an actual microscopic matrix element. If $|s\rangle$ has i down and j up and $|s'\rangle = S_i^+ S_j^- |s\rangle$, then

$$\langle s'|H(\boldsymbol{\theta})|s\rangle = -J_\pm e^{+2i\boldsymbol{\theta}\cdot\boldsymbol{d}_{ij}}, \qquad \langle s|H(\boldsymbol{\theta})|s'\rangle = \langle s'|H(\boldsymbol{\theta})|s\rangle^*. \tag{11}$$

Equation (10) is Hermitian for every $\boldsymbol{\theta}$ and reduces to the original Hamiltonian at $\boldsymbol{\theta} = 0$. The source is a transport-counting twist, not a fitted field or counterterm. Everything up to this point is an exact finite-cluster operator construction: no low-energy reduction, energy denominator, or expansion in either transverse coupling has been introduced.

To see what it counts, consider a sequence γ of microscopic moves. Step ℓ acts on the stored oriented bond (i_ℓ, j_ℓ). Define

$$\Delta\boldsymbol{p}_\ell = \begin{cases} -\boldsymbol{d}_{ij}, & T_\ell = S_i^+ S_j^-, \\ +\boldsymbol{d}_{ij}, & T_\ell = S_i^- S_j^+, \\ +\boldsymbol{c}_{ij}, & T_\ell = S_i^+ S_j^+, \\ -\boldsymbol{c}_{ij}, & T_\ell = S_i^- S_j^-. \end{cases} \tag{12}$$

Multiplication of the microscopic matrix elements gives

$$\prod_{\ell\in\gamma} e^{-2i\boldsymbol{\theta}\cdot\Delta\boldsymbol{p}_\ell} = e^{-i\boldsymbol{\theta}\cdot\boldsymbol{q}_\gamma}, \qquad \boldsymbol{q}_\gamma = 2\sum_{\ell\in\gamma} \Delta\boldsymbol{p}_\ell. \tag{13}$$

The coordinate normalization is chosen so that a completed path between ice-rule configurations has $\boldsymbol{q}_\gamma \in \mathbb{Z}^3$. A contractible closed path has $\boldsymbol{q}_\gamma = 0$; a path closed only by periodic identification has nonzero $\boldsymbol{q}_\gamma$. Changing home-cell representatives changes bond increments and endpoint phases together, leaving the completed $\boldsymbol{q}_\gamma$ and especially the condition $\boldsymbol{q}_\gamma = 0$ invariant. Equation (13) is likewise an exact product of microscopic matrix-element phases. Calling γ a path does not make this step perturbative.

For integer completed transport, Fourier orthogonality gives

$$\frac{1}{(2\pi)^3} \int d^3\theta\, e^{-i\boldsymbol{\theta}\cdot\boldsymbol{q}_\gamma} = \delta_{\boldsymbol{q}_\gamma,\boldsymbol{0}}. \tag{14}$$

As a reproducible cubic-16 example, the sequence $S_0^- S_1^+$, $S_8^+ S_{10}^+$, $S_8^- S_{11}^-$ has increments

$$(0, 1/4, 1/4), \quad (5/4, 0, 5/4), \quad (-5/4, -1/4, -1), \tag{15}$$

and hence $\boldsymbol{q}_\gamma = (0, 0, 1)$. It connects two ice configurations but is a spurious $J_\pm J_{\pm\pm}^2$ four-spin process, so Eq. (14) annihilates it. A pair flip followed by its inverse instead has increments $-\boldsymbol{c}_{ij}$ and $+\boldsymbol{c}_{ij}$ and is retained because its total transport vanishes.

The averaging is performed only after exact band extraction and canonical pullback. Averaging $H(\boldsymbol{\theta})$ directly would erase individual moves before they can compose. Averaging $h(\boldsymbol{\theta})$ instead removes its nonzero-transport Fourier coefficients while retaining all zero-transport paths assembled from those same microscopic moves.

The corresponding code operation is exactly

```
d = r[j] - r[i] - image_ij @ L
if spin[i] != spin[j]:
    phase = exp(2j * theta @ d)
    amplitude = -Jpm * (phase if spin[i] == down else conjugate(phase))
else:
    c = r[i] + r[j] - image_ij @ L
    phase = exp(-2j * theta @ c)
    amplitude = Jpmpm * (phase if spin[i] == down else conjugate(phase))
H[flip_both(i, j, state), state] += amplitude
```

Because $J_{\pm\pm}$ changes total S^z by two, a finite-$J_{\pm\pm}$ calculation uses the full 2^N spin basis (or another basis explicitly closed under pair flips). Cubic-16 has dimension 65,536 in this representation.

3 The exact source-dependent band projector

The notation $P_{\boldsymbol{\theta}}$ does not denote the ice projector and does not insert the source. It is the spectral projector of the already sourced microscopic Hamiltonian $H(\boldsymbol{\theta})$ onto one isolated band.

More precisely, turn on the transverse exchange continuously in $H_s(\boldsymbol{\theta}) = H_0 + s[H(\boldsymbol{\theta}) - H_0]$, $0 \le s \le 1$. Assume the band issuing from the ice-rule subspace remains separated from the rest of the spectrum. If $\Gamma_{\boldsymbol{\theta}}$ encloses that band and no other eigenvalues at $s = 1$, define the Riesz projector

$$P_{\boldsymbol{\theta}} = \frac{1}{2\pi i}\oint_{\Gamma_{\boldsymbol{\theta}}} (z - H(\boldsymbol{\theta}))^{-1}\, dz = \sum_{\alpha \in \mathcal{B}_{\boldsymbol{\theta}}} |\psi_{\alpha\boldsymbol{\theta}}\rangle\langle\psi_{\alpha\boldsymbol{\theta}}|. \tag{16}$$

By construction, $\operatorname{rank} P_{\boldsymbol{\theta}} = \operatorname{rank} P_{\text{ice}}$. A gap closing makes the continuation ambiguous and is a failure of the protocol.

Different diagonalizations choose different unitary mixtures inside the same band. The polar map removes that arbitrary gauge. On the ice-rule subspace, define

$$G_{\boldsymbol{\theta}} = P_{\text{ice}} P_{\boldsymbol{\theta}} P_{\text{ice}}, \qquad W_{\boldsymbol{\theta}} = P_{\boldsymbol{\theta}} P_{\text{ice}} G_{\boldsymbol{\theta}}^{-1/2}. \tag{17}$$

When $G_{\boldsymbol{\theta}}$ is positive definite, $W_{\boldsymbol{\theta}}^\dagger W_{\boldsymbol{\theta}} = P_{\text{ice}}$ and $W_{\boldsymbol{\theta}} W_{\boldsymbol{\theta}}^\dagger = P_{\boldsymbol{\theta}}$. The exact band operator in one fixed reference space is

$$h(\boldsymbol{\theta}) = W_{\boldsymbol{\theta}}^\dagger H(\boldsymbol{\theta}) W_{\boldsymbol{\theta}}. \tag{18}$$

A singular $G_{\boldsymbol{\theta}}$ means that the exact band can no longer be canonically identified with the ice-rule subspace; it must not be repaired by an arbitrary regularizer.

Numerically, let B be an isometry whose columns span the ice-rule subspace, let $\Psi_{\boldsymbol{\theta}}$ contain orthonormal exact band eigenvectors, and let $E_{\boldsymbol{\theta}}$ be their eigenvalues. With

$$X_{\boldsymbol{\theta}} = B^\dagger \Psi_{\boldsymbol{\theta}}, \qquad U_{\boldsymbol{\theta}} = X_{\boldsymbol{\theta}} (X_{\boldsymbol{\theta}}^\dagger X_{\boldsymbol{\theta}})^{-1/2}, \tag{19}$$

the matrix actually averaged by the code is

$$h(\boldsymbol{\theta}) = U_{\boldsymbol{\theta}}\, \operatorname{diag}(E_{\boldsymbol{\theta}})\, U_{\boldsymbol{\theta}}^\dagger. \tag{20}$$

The smallest eigenvalue of $X_{\boldsymbol{\theta}}^\dagger X_{\boldsymbol{\theta}}$ is the reported band-overlap diagnostic.

4 Zero-character projection

Because completed transport is integer valued, the pulled-back band has a Fourier decomposition

$$h(\boldsymbol{\theta}) = \sum_{\boldsymbol{q}\in\mathbb{Z}^3} h_{\boldsymbol{q}} e^{-i\boldsymbol{\theta}\cdot\boldsymbol{q}}. \tag{21}$$

The desired operator is the zero Fourier coefficient,

$$h_{\boldsymbol{q}=0} = \frac{1}{(2\pi)^3}\int_{[0,2\pi)^3} d^3\theta\, h(\boldsymbol{\theta}). \tag{22}$$

The implementation approximates this integral by the complete character grid

$$h_M = \frac{1}{M^3}\sum_{\boldsymbol{m}\in\mathbb{Z}_M^3} h\left(\frac{2\pi}{M}\boldsymbol{m}\right). \tag{23}$$

Discrete Fourier orthogonality gives

$$h_M = \sum_{\boldsymbol{q}\in M\mathbb{Z}^3} h_{\boldsymbol{q}}. \tag{24}$$

Finite M therefore aliases transports $M\boldsymbol{k}$ with zero transport. It is not an exact all-order selector. The controlled numerical check is convergence of h_M as M is increased.

The order of operations matters. One must first construct $H(\boldsymbol{\theta})$, diagonalize it, form $P_{\boldsymbol{\theta}}$, and pull the band back before averaging. Averaging $H(\boldsymbol{\theta})$ first would remove individual exchanges before they can compose. Averaging $\operatorname{Tr} e^{-\beta H(\boldsymbol{\theta})}$ would average a nonlinear observable rather than define $h_{\boldsymbol{q}=0}$.

5 In what sense is the construction nonperturbative?

At every source point the production calculation diagonalizes the microscopic $H(\boldsymbol{\theta})$ at the target values of $(J_\pm, J_{\pm\pm})/J_{zz}$. It does not expand the eigenvalues, $P_{\boldsymbol{\theta}}$, $G_{\boldsymbol{\theta}}^{-1/2}$, or $h(\boldsymbol{\theta})$ in that ratio. The Riesz projector and polar map therefore contain repeated virtual excursions out of the ice-rule subspace to all orders allowed by the finite Hilbert space.

Perturbation theory provides a useful identity, not the numerical definition. If a formal coupling λ is introduced and the isolated band is analytic, then

$$h(\boldsymbol{\theta};\lambda) = \sum_{n=0}^{\infty}\lambda^n \sum_{\boldsymbol{q}\in\mathbb{Z}^3} h_{n,\boldsymbol{q}} e^{-i\boldsymbol{\theta}\cdot\boldsymbol{q}}. \tag{25}$$

Substitution into Eq. (22) yields

$$h_{\boldsymbol{q}=0}(\lambda) = \sum_{n=0}^{\infty}\lambda^n h_{n,0}. \tag{26}$$

Low-order Schrieffer–Wolff theory computes a few terms on the right. The production protocol instead evaluates the left-hand side directly at $\lambda = 1$. This is the precise nonperturbative statement.

It is also a limited statement. The result is exact only for the chosen finite-cluster band, up to eigensolver tolerance and the $M \to \infty$ quadrature limit. It requires band isolation and a nonsingular polar overlap. It does not establish cluster-size convergence or bulk QSI physics.

6 Production algorithm

The implementation follows Eqs. (10)–(23):

```
B = ice_projector_basis()  # B @ B.H = P_ice

for theta in complete_character_grid(M):
    Htheta = microscopic_character_hamiltonian(theta)
    Etheta, Psitheta = isolated_band_connected_to_P_ice(Htheta)
    Xtheta = B.H @ Psitheta
    require min_eig(Xtheta.H @ Xtheta) > overlap_cutoff
    Utheta = Xtheta @ inverse_sqrt(Xtheta.H @ Xtheta)
    htheta[theta] = Utheta @ diag(Etheta) @ Utheta.H

hM = mean(htheta over the complete grid)
compare centered(hM) with centered(h_previous_M)
```

The current implementation works in a conserved magnetization sector and uses translation sectors at zero source as an acceleration. These symmetry choices do not alter the definitions above.

7 Low-order path enumeration is an audit

The path code is retained to prove that the source distinguishes the known finite-torus loop from a contractible hexagon. Let $Q_{\text{ice}} = 1 - P_{\text{ice}}$ and

$$R = -Q_{\text{ice}}(H_0 - E_{\text{ice}})^{-1}Q_{\text{ice}}. \tag{27}$$

Because $P_{\text{ice}}VP_{\text{ice}} = 0$ in the XXZ model, the first diagnostic blocks are

$$h^{(2)} = P_{\text{ice}}VRVP_{\text{ice}}, \tag{28}$$

$$h^{(3)} = P_{\text{ice}}VRVRVP_{\text{ice}}. \tag{29}$$

Each row remains keyed by its source state, target state, and accumulated image vector until $\boldsymbol{q}_\gamma$ is evaluated. Equation (13) then verifies directly that winding four-loops have nonzero transport while contractible hexagons have zero transport. These rows test the meaning of the source; they do not generate the production spectrum. In terms of absolute row weight, the $\boldsymbol{q} = 0$ selector retains 0% of the winding four-loop and wrapping-hexagon channels and 100% of the contractible-hexagon channel on both clusters.

The geometry census used by this audit is listed in Table 1. Neither cluster contains a contractible

Table 1: Finite-cluster loop census used by the low-order topology audit.

cluster	winding 4	contractible 6	wrapping 6
cubic-16	36	16	48
FCC-32	24	32	96

four-cycle.

8 Full-Hilbert Hamiltonian and thermodynamics

The pulled-back band is an intermediate construction that determines the counterterm. It is not the state space used for thermodynamics: final observables use one Hermitian operator on the complete microscopic Hilbert space. Define

$$\bar{h}_M = h_M + \frac{\mathrm{Tr}[h(\mathbf{0}) - h_M]}{d} I_d, \qquad d = \mathrm{rank}\, P_{\mathrm{ice}}, \tag{30}$$

and embed its difference from the periodic band with the exact zero-source isometry:

$$H_{\mathrm{wf}}^{(M)} = H(\mathbf{0}) + W_0[\bar{h}_M - h(\mathbf{0})]W_0^\dagger. \tag{31}$$

Because $W_0 W_0^\dagger = P_0$, this full-Hilbert Hamiltonian equals $H(\mathbf{0})$ on the exact band complement. Its correction is generally nonlocal, but no state outside the selected band is discarded. The partition function is the ordinary trace $Z_{\mathrm{wf}} = \mathrm{Tr}_{\mathcal{H}}\, e^{-\beta H_{\mathrm{wf}}^{(M)}}$. Replacing the lowest d eigenvalues in the stored spectrum is merely an efficient evaluation of this operator identity.

For any finite spectrum, the thermodynamics are evaluated after shifting by the ground-state energy to stabilize the Boltzmann weights:

$$Z' = \sum_i e^{-\beta(E_i - E_0)}, \tag{32}$$

$$\langle E \rangle = E_0 + \frac{\sum_i (E_i - E_0) e^{-\beta(E_i - E_0)}}{Z'}, \tag{33}$$

$$C = \frac{\langle (E - E_0)^2 \rangle - \langle E - E_0 \rangle^2}{T^2}, \tag{34}$$

$$S = \ln Z' + \beta \langle E - E_0 \rangle. \tag{35}$$

All extensive quantities are divided by the number of sites. The temperature grid is logarithmic from $10^{-4} J_{zz}$ to $2J_{zz}$ with 1400 points.

9 External verification against Huang et al.

Huang, Deng, Wan, and Meng simulate Eq. (1) at the sign-free point $(J_\pm, J_{\pm\pm})/J_{zz} = (0.046, 0)$ using continuous-time worm QMC [1]. This is an especially useful reference because its sign and coupling convention match the repository directly.

The heat-capacity and entropy curves are extracted from the original vector paths in Fig. 1(b) of the arXiv source. The extractor records explicit axis calibrations and writes a CSV with estimated digitization uncertainties. The comparison metric interpolates the finite-cluster curve on $\log T$ and then computes an RMSE over the reference points. Peak gates use fixed temperature windows and log-position errors.

The current numerical summary is given in Table 2. The converged $M = 4$ low-temperature heat-capacity RMSE is reduced by 72.9%, but 0.02327 remains above the threshold 0.02. Its entropy RMSE 0.04509 also fails. Removing winding transfer does not guarantee correct finite-cluster state counting or degeneracy, so these failures diagnose the cluster rather than a lack of character convergence.

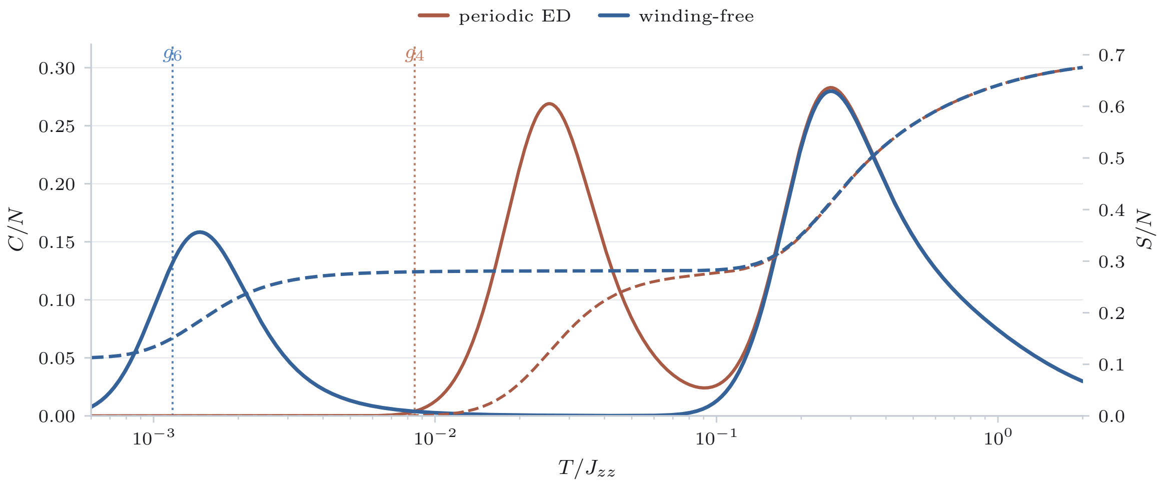

Figure 2: Periodic and winding-free thermodynamics at $(J_\pm, J_{\pm\pm})/J_{zz} = (0.046, 0)$. Solid curves and the left axis show C/N; dashed curves and the right axis show S/N. The exact $M = 3, 4$ centered operators differ by 0.013%.

Table 2: Thermodynamic comparison with the Huang *et al.* QMC data.

quantity	periodic cubic-16	winding-free cubic-16	QMC
low T_p/J_{zz}	0.02554	0.001463	0.001100
low-T C/N RMSE	0.0859	0.02327	0
high T_p/J_{zz}	0.2549	0.2549	0.2105
S/N RMSE	0.0514	0.04509	0

10 Internal winding-free dynamics

The longitudinal comparison holds the physical probe fixed. With W_0 the exact zero-source canonical map, define

$$\widetilde{S}^z_a(\boldsymbol{q}) = W_0^\dagger \left(\sum_{j\in a} e^{-2\pi i \boldsymbol{q}\cdot\boldsymbol{r}_j} S^z_j \right) W_0. \tag{36}$$

The same operator is evaluated with the periodic Hamiltonian $h(\mathbf{0})$ and the winding-free Hamiltonian $h_{M=4}$. Thus the comparison changes the band Hamiltonian, not the probe convention. Cubic-16 supports only Γ and the three symmetry-related X momenta.

The campaign script reconstructs W_0 from the full fixed-S^z eigenvectors, checks its isometry, verifies the cached periodic operator, pulls back the sixteen microscopic S^z_i operators, and then evaluates the exact Lehmann sum in each 90-dimensional band Hamiltonian. Gaussian broadening is $6\times10^{-4}J_{zz}$. The saved output contains both unbroadened transition lines and broadened spectra.

The same QMC paper supplies a future dynamical benchmark from stochastic analytic continuation. It reports the sublattice traces

$$S^{zz}(\boldsymbol{q},\omega) = \sum_{\alpha=1}^{4} S^{zz}_{\alpha\alpha}(\boldsymbol{q},\omega), \qquad S^{+-}(\boldsymbol{q},\omega) = \sum_{\alpha=1}^{4} S^{+-}_{\alpha\alpha}(\boldsymbol{q},\omega) \tag{37}$$

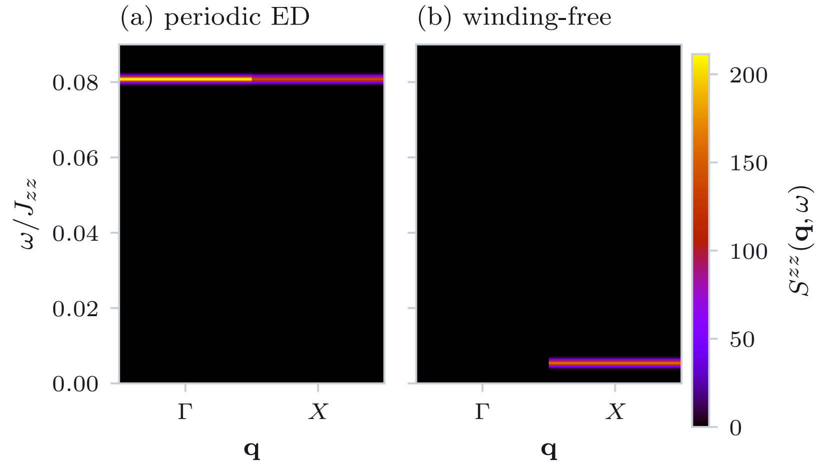

Figure 3: Periodic and winding-free four-sublattice-traced $S^{zz}(\boldsymbol{q},\omega)$ on common linear scales. The periodic line begins at $0.08065J_{zz}$; winding-free weight is suppressed to 2×10^{-6} at Γ and lies at $0.005362J_{zz}$ at each X point.

at $T/J_{zz} = 0.001$, 0.04, and 0.1. A valid ED comparison must use only momenta in the finite cluster's translation group, match the sublattice trace and spin normalization, use a linear frequency axis, and state its broadening. This dynamics gate remains pending.

11 How the campaign runner fits together

The production data path is short enough to audit directly:

```
T = geomspace(1e-4, 2.0, 1400)

d = rank(P_ice)
E0, Psi0 = exact_band_connected_to_P_ice(H(theta=0), count=d)
h0 = polar_pullback(E0, Psi0, P_ice_basis)
h2 = average(exact_gauge_orbit(h0, M=2))

for M in [3, 4]:
    for conjugation representative theta in Z_M^3:
        E, Psi = eigsh(H(theta), lowest=d+1)
        verify gap, Ritz residual, and ice overlap
        h[theta] = polar_pullback(E[:d], Psi[:, :d], P_ice_basis)
        h[-theta] = conjugate(h[theta])
    hM = average(h over Z_M^3)

E_full = load cached exact cubic16 spectrum
E_wf = spectrum_of_full_Hwf(E_full, eigvalsh(hM), trace_match=True)
compute C/N and S/N from E_full and E_wf

load Huang vector-extracted C/N and S/N
evaluate fixed gates
write nonperturbative operators, curves, caches, and report
```

No fitting parameter is adjusted in this sequence. The model coupling is fixed to the QMC value and no counterterm or perturbative coefficient enters the production operator.

12 Code map

Table 3: Repository components used by the winding-free calculation.

file or function	responsibility
`notes/recompute_finite_size_artifact.py`	active geometry, ice basis, cycle census, virtual hopping, order-two/order-three row tables, transported dipole, and dense assembly
`build_cluster`	positions, bonds, image vectors, tetrahedra, loops, and ice states
`apply_v`	spin exchange, amplitude sign, and image-transport update
`merge_rows`	coherent sum at fixed source, target, and image vector
`sw_order23`	explicit $PVRVP$ and $PVRVRVP$ row construction
`transport_delta`	integer completed-path transport in Eq. (13)
`assemble`	periodic or direct winding-free order-three matrix
`src/qsi_campaign/protocol.py`	polar pullback, character average, centered operator error, and full-Hilbert counterterm embedding
`src/qsi_campaign/exact_band.py`	fixed-magnetization basis, microscopic character Hamiltonian, sparse exact-band extraction, gauge orbit, and overlap/Ritz diagnostics
`src/qsi_campaign/thermodynamics.py`	stable E/N, C/N, and S/N from a finite spectrum
`src/qsi_campaign/benchmarks.py`	vector-data loading and log-grid RMSE
`campaign/data/extract_huang2018.py`	reproducible vector extraction of Huang et al. Fig. 1(b)
`campaign/run_nonperturbative.py`	active exact $M = 2, 3, 4$ campaign, restartable point cache, thermodynamics, and gate report
`campaign/run_validation.py`	perturbative topology and cluster-size diagnostic
`campaign/make_figures.py`	figures generated only from saved campaign products

13 Current gates and their interpretation

All active grids use the same primitive $\boldsymbol{q} = 2\boldsymbol{\delta}$ source. The large $M = 1 \to 2$ and 6.36% $M = 2 \to 3$ changes show why a single parity mask is not enough; the 0.013% $M = 3 \to 4$ change supplies the first passed exact-band character gate.

Table 4: Current validation gates and their interpretation.

gate	status	interpretation
loop topology	pass	winding four-loops removed; contractible hexagons retained
internal S^{zz} dynamics	pass	common W_0-dressed probe used for periodic and winding-free bands
all-temperature stability	pass	high-T peak position unchanged on cubic-16
QMC heat capacity	fail	exact $M = 4$ low-T RMSE $0.02327 > 0.02$
QMC entropy	fail	exact $M = 4$ RMSE $0.04509 > 0.02$
fixed-S^z band gap	pass	minimum source-resolved gap exceeds $0.627J_{zz}$
band overlap	pass	minimum exact-band ice overlap is 0.762
character resolution	pass	centered $M = 3 \to 4$ error is 0.013%
order convergence	diagnostic only	order 2/3 is not the production method
QMC-SAC dynamics	pending	matched S^{zz} and S^{+-} not computed
FCC-32 full Hamiltonian	pending	current FCC-32 result is ice-manifold order three

14 Common implementation mistakes

1. **Discarding image transport too early.** Diagnostic rows must remain keyed by source, target, and image vector until after transport is evaluated.
2. **Calling the order-three block "exact ED."** Its diagonalization is exact only for that truncated operator on the ice-rule subspace.
3. **Calling FCC-32 a full-Hamiltonian result.** The active FCC-32 spectrum contains only the order-three ice block; the 2^{32} complement is absent.
4. **Treating finite M as an exact all-order projection.** Finite characters select $\boldsymbol{q} = 0 \pmod{M}$, not only $\boldsymbol{q} = 0$.
5. **Tracking eigenvalues without eigenvectors.** Exact-band identity requires spectral separation and a nonsingular projected Gram matrix.
6. **Averaging thermal observables.** Character projection is linear at the operator level; $e^{-\beta H}$ is nonlinear.
7. **Comparing dynamics at arbitrary momenta.** A finite cluster has a discrete reciprocal translation group. Plot interpolation cannot create missing physical momenta.
8. **Using a heat-capacity peak as the only gate.** The present entropy failure already shows why curve shape and state counting matter.

15 Reproduction and extension checklist

From the repository root, the active workflow is

```
python -m pip install -e .
make all
make nonperturbative-m4
```

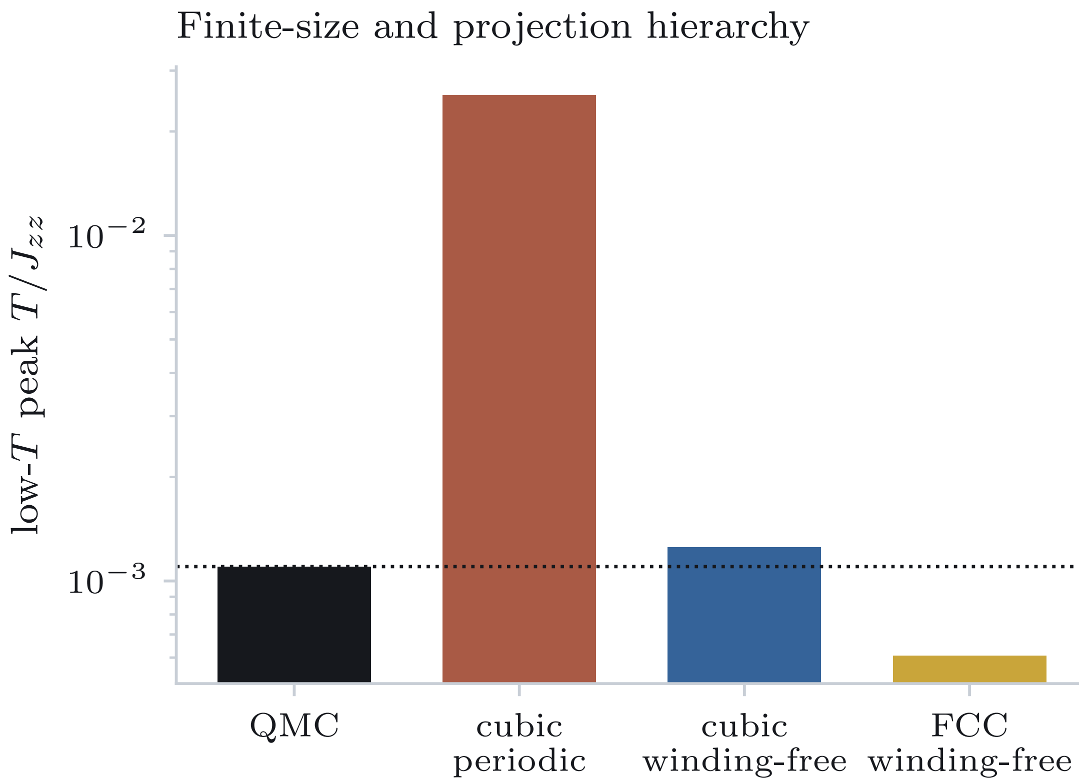

Figure 4: Validation hierarchy. Topological correctness at orders two and three is necessary but not sufficient; the external and convergence gates must be passed before material-level claims are made.

It regenerates the row-resolved campaign, runs the unit tests, builds figures, and compiles the manuscript, Supplemental Material, and this note. The machine-readable exact results are `campaign/outputs/nonperturbative_cubic16_p0p046000.npz` and its JSON report. Individual source points are cached under `campaign/cache/nonperturbative_points/`.

An extension should proceed in this order:

1. optionally extend to $M = 5$ as a tighter character check;
2. include all conserved magnetization sectors in the unrestricted gap;
3. diagnose the heat and entropy failures before interpreting the low-temperature peak as bulk convergence;
4. match the QMC-SAC channels at the finite cluster's actual momenta; and
5. only then deploy a stochastic-complement method on FCC-32 and proceed to a material fit.

16 Takeaway

The protocol's useful idea is simple: a periodic image label is physical bookkeeping for a virtual path, even though it is invisible in the wrapped spin configuration. Keeping that label until a completed path returns to the ice band makes winding and contractible processes separable. The present implementation proves the path classification at orders two and three and then executes the production projection with exact microscopic eigenspaces. The $M = 3, 4$ agreement removes perturbative-order ambiguity from cubic-16. The remaining heat and entropy failures show that this converged finite cluster still cannot support a bulk or material claim.

References

[1] Chun-Jiong Huang, Youjin Deng, Yuan Wan, and Zi Yang Meng. Dynamics of topological excitations in a model quantum spin ice. *Phys. Rev. Lett.*, 120:167202, 2018.